

aiida-aimall: A Python package for automating workflows for AIMAll software

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Summary

Since its introduction by Richard Bader, the Quantum Theory of Atoms in Molecules (QTAIM) has become a useful tool for computational chemists. This Python package provides plugins for a common QTAIM software, AIMAll, for the AiiDA Python infrastructure. aiida-aimall is an essential tool for ensuring reproducible calculations, with full generation history. Workflows are also provided to interface AIMAll software with any quantum chemistry package that can be run through the command line, so long as it generates the input files required by AIMAll.

Statement of need

aiida-aimall is a Python package based on the AiiDA ([Talirz et al., 2020](#)) infrastructure designed to assist users with generating inputs for AIMAll software ([Keith, 2019](#)). The goal of the AiiDA infrastructure is, in part, to ensure data provenance and calculation reproducibility. While aiida-aimall has been developed primarily for interface with Gaussian software outputs ([Frisch et al., 2016](#)), a versatile workflow enabling interface with other quantum chemistry packages is also made available. To the best of the authors' knowledge, no tool for automatically linking the output of quantum chemistry packages with AIMAll while tracking data provenance exists. This tool significantly simplifies complex multi-step workflows such as substituent parameter generation into one simple step, easing the burden on the end user and allowing for high-throughput calculations.

Through a variety of workflows that can start with an .xyz file, AiiDA StructureData, or even with a SMILES string of a molecule, aiida-aimall provides a variety of use cases for automating complex workflows. Additionally, tools to ensure that computers are not overloaded through too many simultaneous processes are made available through classes of FromGroupSubmissionControllers from aiida-submission-controller to limit active processes.

Features

aiida-aimall contains many different classes from aiida tailored to ensure ease of use of AIMAll calculations. Numerous features provided by aiida-aimall are described in full on the [documentation webpage hosted on ReadTheDocs](#). A brief description of main features is provided here. For each of the types of workflows and calculations that follow, the inputs, outputs, and all processes (such as calcfuctions or CalcJobs) that occurred in generating the results are added to a database to provide a record of the full calculation history and ensure reproducibility.

Running Simple AIMAll Calculations

The simplest functionality provided by `aiida-aimall` is running AIMAll calculations. All AIMAll calculations utilize the `AimqbParameters` datatype provided by `aiida-aimall`. The `AimqbParameters` datatype is a validator for AIMAll command line input. Command line parameters are to be provided as a dictionary, then `AimqbParameters` ensures that the parameters match options available for AIMAll software as [defined on the software website](#), and that the correct data type is provided for each parameter. In this way, `AimqbParameters` verifies the provided input to AIMAll calculations prior to launch of the calculation. These parameters, along with `SinglefileData` of a valid AIMAll input file, a `Code` object for AIMAll software, and relevant metadata are provided to an `AimqbCalculation`.

This functionality in itself is an overcomplication of the simple process of running the software normally. However, it does have some benefits. The output is already extracted and stored in the database in a readily usable manner through the use of the `AimqbBaseParser`. It is now simple to see the history of the calculation.

Substituent Properties

Some of the workflows in `aiida-aimall` automate calculation of substituent properties from AIMAll output. These substituent properties have been developed by the authors. ([Lefrancois-Gagnon & Mawhinney, 2023](#)) The `SubstituentParameterWorkChain` does this automatically, and any routine AIMAll calculation can make use of this by using the `AimqbGroupParser`, which can be provided in metadata input to `AimqbCalculation` as an entry in the metadata dictionary: `metadata.options.parser_name: 'aimall.group'`. A detailed description of the calculated substituent properties is available [in a tutorial in the documentation](#). AIMAll integrated and graph properties are obtained.

Integrations with Computational Chemistry Software

`aiida-aimall`'s main draw is that it enables automation to link the outputs of standard computational chemistry software directly to an AIMAll calculation. A list of provided workflows is shown in Table 1. The software with the most robust implementation is Gaussian software, ([Frisch et al., 2016](#)) as Gaussian already has an implemented `aiida` package. Other computational chemistry software like ORCA can be run through the `QMToAIMWorkchain`, which uses `aiida-shell` to run software than can be run through the command line. These workflow inputs, outputs and processes are illustrated in the provenance graph in [Figure 1](#). If `.molden` or `.cp2k.out` output formats are available, one could alternatively use these to generate the needed `.wfx` files for AIMAll, and automatically run AIMAll through the `GenerateWFXToAIMWorkchain`.

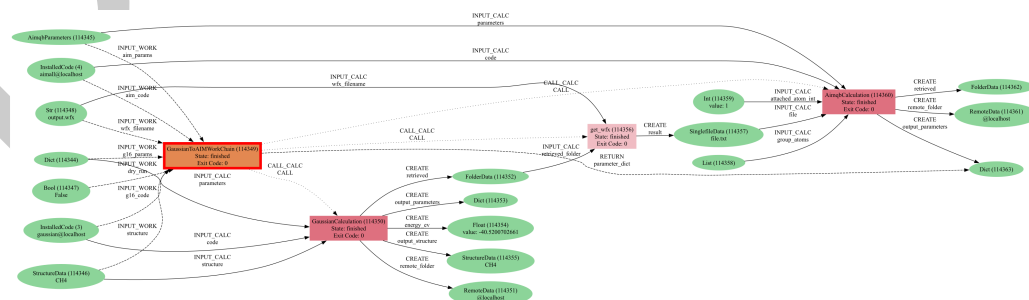


Figure 1: Database inputs, outputs, and process of the GaussianToAimWorkchain.

Table 1: Main workflows provided by `aiida-aimall`, their `aiida` entry points that can be used to load them by `aiida.plugins.WorkflowFactory`, and a brief description. These workflows all end with the output of an `AimqbCalculation` as their main output.

```

73 |-----+-----+-----+-----+ | Workflow | Entry
74 Point | Purpose | +-----+-----+-----+-----+ |
75 QMToAIMWorkchain | aimall.qmtoaim | Run a general computational ||| chemistry software and
76 link ||| it to an AIMAll ||| calculation | +-----+-----+-----+
77 | GenerateWFXToAIMWorkchain | aimall.wfxtoaim | Take non-standard AIMAll ||| input
78 files, and run AIMAll | +-----+-----+-----+
79 | GaussianToAIMWorkChain | aimall.g16toaim | Run a Gaussian calculation |||
80 | and automatically run ||| an AIMAll calculation on ||| its outputs |
81 +-----+-----+-----+-----+ | SubstituentParameterWorkChain
82 | aimall.subparam | Compute substituent ||| properties defined by ||| the authors
83 automatically | +-----+-----+-----+-----+

```

84 Controllers to limit computer burden when running large numbers of jobs

85 The last main contribution of aiida-aimall is through the definition of controllers from
 86 the aiida-submission-controller package. These controllers limit active processes and
 87 can be used together as demonstrated in a [tutorial notebook](#) to automate the entire
 88 SubstituentParameterWorkchain. These use a number of Workchains developed just
 89 for their use in these controllers. The process flows as SmilesToGaussianController ->
 90 AIMAllReorController -> GaussianController -> AIMAllController. The latter two con-
 91 trollers can also be seen and used as general use controllers wrapping GaussianCalculations
 92 and AimqbCalculations

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